

Assignment 1

Lab Report: Human Glucagon (GCG)

Adnan Bhatt
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Introduction

In this report, we analyze the human Glucagon (GCG) gene and protein. The primary objective is to examine the FASTA sequence, derive the corresponding amino acid sequence, and discuss the secondary structures and gene expression profile of this protein. Glucagon is a peptide hormone that plays a crucial role in maintaining blood glucose levels. It is secreted by pancreatic alpha cells and acts opposite to insulin by stimulating glycogen breakdown and gluconeogenesis in the liver. Understanding its sequence and structure helps explain its biological function and relevance in metabolic disorders such as diabetes mellitus.

FASTA Sequence of Human Glucagon

The FASTA sequence for the human glucagon protein is obtained from the NCBI website.

- Gene Name: GCG
- Organism: Homo sapiens
- Gene ID: 2641
- Chromosome Location: 2q24.2
- Protein Length (Preproglucagon): 180 amino acids

Below is the FASTA sequence for the human glucagon protein:

```
>NM_002054.5 Homo sapiens glucagon (GCG), mRNA
ACAGAGCTTAGGACACAGAGCACATCAAAAGTTCCCAAAGAGGGGCTTGCTCTCTCTTCACCTGCTCTGTT
CTACAGCACA CTACCAGAAGACAGCAGAAATGAAAAGCATT TACTTTGTGGCTGGATTATTTGTAATGCT
GGTACAAGGCAGCTGGCAACGTTCCCTTCAAGACACAGAGGAGAAATCCAGATCATTCTCAGCTTCCCAG
GCAGACCCACTCAGTGATCCTGATCAGATGAACGAGGACAAGCGCCATT CACAGGGCACATT CACCAGTG
ACTACAGCAAGTATCTGGACTCCAGGCGTGCCCAAGATTTTGTGCAGTGGTTGATGAATACCAAGAGGAA
CAGGAATAACATTGCCAAACGTCACGATGAATTTGAGAGACATGCTGAAGGGACCTTTACCAGTGATGTA
AGTTCTTATTTGGAAGGCCAAGCTGCCAAGGAATTCATTGCTTGGCTGGTGAAAGGCCGAGGAAGGCGAG
ATTTCCCAGAAGAGGTCGCCATTGTTGAAGAACTTGGCCGCAGACATGCTGATGGTTCTTTCTCTGATGA
GATGAACACCATTCTTGATAATCTTGCCGCCAGGGACTTTATAAACTGGTTGATT CAGACCAAAATCACT
GACAGGAAATAACTATATCACTATTCAAGATCATCTTCACAACATCACCTGCTAGCCACGTGGGATGTTT
GAAATGTTAAGTCCTGTAAATTTAAGAGGTGTATTCTGAGGCCACATTGCTTTGCATGCCAATAAATAAAA
TTTTCTTTTAGTGTTGTGTAGCCAAAAATTACAAATGGAATAAAGTTTTATCAAAATATTGCTAAAATAT
CAGCTTTAAAAATATGAAAGTGCTAGATTCTGTTATTTTCTTCTTATTTTGGATGAAGTACCCCAACCTGT
TTACATTTAGCGATAAAATTATTTTTCTATGATATAATTTGTAAATGTAAATTATTCCGATCTGACATAT
CTGCATTATAATAATAGGAGAATAGAAGAACTGGTAGCCACAGTGGTGAAATTGGAAAGAGAACTTTCTT
CCTGAAACCTTTGTCTTAAAAATACTCAGCTTTCAATGTATCAAAGATACAATTAAATAAAATTTTCAAG
CTTC
```

Preproglucagon Amino Acid Sequence (180 Amino Acids)

The human GCG gene encodes a precursor protein called preproglucagon, which consists of 180 amino acids. The complete amino acid sequence obtained from NCBI (RefSeq: NP_002045.1) is shown below:

```
>NP_002045.1 pro-glucagon preproprotein [Homo sapiens]
MKSIYFVAGLFVMLVQGSWQ RSLQDTEEKSRFSASQADPLSDPDQMNE DKRHSQGTFTSDYSKYLDSRR
AQDFVQWLMNTKRN RNNIAKRHDEFERHAEGTFTSDVSSYLEGQAAKEFIAWLVKGRGR RDPPEEVAIVE
ELGRRHADGSFSDEMNTILDNLAARDFINWLIQTKITDRK
```

This precursor protein undergoes post-translational processing to produce biologically active peptides including glucagon (29 amino acids), GLP-1, GLP-2, and oxyntomodulin.

Mature Glucagon Peptide (29 Amino Acids)

```
HSQGTFTSDYSKYLDSRR AQDFVQWLMNT
```

Amino Acid Composition and Properties

Position	1-Letter	3-Letter	Name	Properties
1	H	His	Histidine	Polar, positively charged (basic)
2	S	Ser	Serine	Polar, uncharged, hydrogen bonding
3	Q	Gln	Glutamine	Polar, uncharged
4	G	Gly	Glycine	Non polar, smallest amino acid
5	T	Thr	Threonine	Polar, uncharged, phosphorylation site
6	F	Phe	Phenylalanine	Non polar, aromatic
7	T	Thr	Threonine	Polar, uncharged
8	S	Ser	Serine	Polar, uncharged
9	D	Asp	Aspartic Acid	Polar, negatively charged (acidic)
10	Y	Tyr	Tyrosine	Polar, aromatic, signaling role
11	S	Ser	Serine	Polar, uncharged
12	K	Lys	Lysine	Polar, positively charged (basic)
13	Y	Tyr	Tyrosine	Polar, aromatic
14	L	Leu	Leucine	Non polar, hydrophobic
15	D	Asp	Aspartic Acid	Polar, negatively charged
16	S	Ser	Serine	Polar, uncharged
17	R	Arg	Arginine	Polar, positively charged (basic)
18	R	Arg	Arginine	Polar, positively charged
19	A	Ala	Alanine	Non polar, hydrophobic
20	Q	Gln	Glutamine	Polar, uncharged
21	D	Asp	Aspartic Acid	Polar, negatively charged
22	F	Phe	Phenylalanine	Non polar, aromatic
23	V	Val	Valine	Non polar, hydrophobic
24	Q	Gln	Glutamine	Polar, uncharged
25	W	Trp	Tryptophan	Non polar, aromatic
26	L	Leu	Leucine	Non polar, hydrophobic
27	M	Met	Methionine	Non polar, contains sulfur
28	N	Asn	Asparagine	Polar, uncharged
29	T	Thr	Threonine	Polar, uncharged

Gene Expression and Tissue Localization

The GCG gene is expressed predominantly in:

- **Pancreas:** Specifically in alpha cells of the islets of Langerhans. This is the primary site of glucagon production.

- **Small Intestine:** Enteroendocrine L cells express the GCG gene, but the precursor is processed differently to produce GLP-1 and GLP-2.
- **Brain:** Low levels of expression have been observed in certain regions.

This tissue-specific expression is essential for glucose homeostasis. Dysregulation of glucagon secretion contributes to hyperglycemia and type 2 diabetes.

Secondary Structure Prediction

The secondary structure of glucagon refers to local folding patterns formed by hydrogen bonds between backbone atoms.

Based on sequence analysis and prediction tools such as PSIPRED:

- Alpha helices: The central region tends to form an alpha helix, particularly upon receptor binding.
- Random coils: The N-terminal and C-terminal regions show flexible characteristics.
- Beta structures: Minimal beta sheet content due to the short peptide length.

Because glucagon is a small peptide of 29 amino acids, it does not form a complex tertiary structure independently. However, it adopts a more stable alpha-helical conformation when interacting with its receptor.

Conclusion

The human glucagon protein encoded by the GCG gene is a key hormone in glucose regulation. Analysis of its FASTA sequence, amino acid composition, tissue expression, and predicted secondary structure provides insight into its biological function.

Understanding glucagon structure and function is important for studying metabolic disorders such as diabetes and for developing therapeutic strategies targeting glucagon signaling pathways.